BIT MANIPULATION

ULTIMATE INTERVIEW PROBLEM BANK

Easy → Medium → Hard • Algorithms + Algorithm-based Questions

JAVA EDITION

Preparation target

Microsoft • Google • Amazon • Qualcomm • Product-based companies

How to use this problem bank

1. Learn the underlying bitwise concept and algorithm first.
2. Implement the technique in Java without copying.
3. Solve the listed problems from basic to advanced variations.
4. For each solution, explain the approach, edge cases, time complexity and space complexity.
5. Treat advanced topics as depth-building material; interview relevance varies by role and company.

Important note: This is a structured syllabus of problem patterns and variations, not a guarantee that every item is a separate or commonly asked interview question. Some entries require constraints to make the stated XOR/bitwise approach valid.

1. Bitwise Fundamentals

1.	★ Check Even or Odd.
2.	★ Check Whether Kth Bit Is Set.
3.	★ Set the Kth Bit.
4.	★ Clear the Kth Bit.
5.	★ Toggle the Kth Bit.
6.	Update the Kth Bit.
7.	★ Count Set Bits.
8.	Count Unset Bits.
9.	Find Rightmost Set Bit.
10.	Find Rightmost Unset Bit.
11.	Remove Rightmost Set Bit.
12.	Isolate Rightmost Set Bit.
13.	Find Position of Rightmost Set Bit.
14.	Check Whether Two Numbers Have Opposite Signs.
15.	Swap Two Numbers Using XOR.

2. Important Bit Manipulation Algorithms

16.	★ Brian Kernighan's Algorithm — count set bits.
17.	★ Minimum Bit Flips to Convert A into B.
18.	Find Numbers Having Exactly K Set Bits.
19.	Count Set Bits from 1 to N.
20.	Find Total Set Bits in a Range.
21.	★ Check Power of Two.
22.	★ Check Power of Four.
23.	Find Next Power of Two.
24.	Find Previous Power of Two.
25.	Find Largest Power of Two Less Than or Equal to N.
26.	Find Smallest Power of Two Greater Than or Equal to N.
27.	Reverse Bits of a 32-bit Integer.
28.	Reverse Bits in a Given Range.
29.	Check Binary Palindrome.
30.	Reverse Bits Using Divide and Conquer.

3. XOR Algorithms and Problems

31. ★ Single Number — all other elements appear twice.
32. ★ Single Number II — other elements appear thrice.
33. ★ Single Number III — two unique elements.
34. ★ Missing Number Using XOR.
35. Find Duplicate Number Using XOR (under suitable constraints).
36. Find Missing and Repeating Numbers.
37. XOR of All Elements in an Array.
38. XOR of Numbers from 1 to N.
39. ★ XOR of a Given Range [L, R].
40. ★ Prefix XOR Array.
41. ★ Range XOR Queries.
42. ★ Subarray XOR Equals K.
43. ★ Count Subarrays with XOR K.
44. ★ Maximum XOR Subarray.
45. ★ Maximum XOR Pair.
46. Maximum XOR of Two Numbers in an Array.
47. Find XOR of All Subsets.
48. Find the Number Occurring Odd Number of Times.

4. XOR Partitioning Algorithm

49. Find Two Non-Repeating Numbers.
50. Find Two Numbers with Odd Frequencies.
51. Partition Array Based on Rightmost Set Bit.
52. Find the Unique Number Among Pairs.
53. Find Unique Number Among Triplicates Using Bit Counting.
54. Find K Numbers with Odd Occurrences (under suitable constraints).

5. Bit Masking Algorithms

55. ★ Generate All Subsets Using Bitmasking.
56. Generate All Subsequences Using Bitmasking.
57. ★ Generate All Combinations of Size K.
58. Count Total Subsets.
59. Check Whether a Subset Exists.
60. Generate All N-Bit Masks.
61. Check Whether One Mask Is a Subset of Another.
62. Perform Union, Intersection and Difference of Masks.
63. Find Complement of a Bitmask.
64. ★ Enumerate All Submasks.
65. ★ Generate Fixed-Popcount Masks Using Gosper's Hack.
66. Generate All Permutations Using Bitmasking.
67. Find Subset with Given Sum Using Bitmasking.

6. Two's Complement and Bitwise Arithmetic

68. Understand Two's Complement.

69. Find Negative of a Number Using Bitwise Operations.

70. ★ Add Two Integers Without Using +.

71. ★ Subtract Two Integers Without Using -.

72. Multiply Two Integers Using Bitwise Operations.

73. Divide Two Integers Without Using Division.

74. Find Maximum of Two Numbers Using Bitwise Operations.

75. Find Minimum of Two Numbers Using Bitwise Operations.

76. Check Integer Overflow.

77. Find Absolute Value Using Bitwise Operations.

7. Gray Code Algorithms

78. ★ Generate N-Bit Gray Code.

79. Convert Binary to Gray Code.

80. Convert Gray Code to Binary.

81. Find Hamming Distance Between Consecutive Gray Codes.

82. Generate Gray Code Sequence Recursively.

8. Hamming Distance and Bit Counting

83. ★ Hamming Distance Between Two Integers.

84. ★ Total Hamming Distance in an Array.

85. Find Number of Bits to Flip.

86. Find Integer with Minimum Hamming Distance.

87. Count Numbers with Exactly K Set Bits.

88. Find Minimum Operations to Make Two Numbers Equal.

9. Binary Trie Algorithms

89. ★ Implement Binary Trie.

90. ★ Maximum XOR of Two Numbers.

91. Maximum XOR With an Element From Array.

92. Maximum XOR With Queries.

93. Find Minimum XOR Pair.

94. Find Maximum XOR Subarray Using Trie.

95. Count Pairs With XOR in a Given Range.

96. Find Kth Maximum XOR Value.

10. XOR Linear Basis

97.	★ Implement XOR Linear Basis.
98.	★ Find Maximum Subset XOR.
99.	Find Minimum Subset XOR.
100.	Count Distinct Subset XOR Values.
101.	Find Kth Smallest Subset XOR.
102.	Check Whether a Target XOR Can Be Formed.
103.	Find Maximum XOR of Any Subset.

11. Bitwise AND / OR Algorithms

104.	Bitwise AND of Numbers in a Range.
105.	Bitwise OR of Numbers in a Range.
106.	★ Count Distinct OR Values of All Subarrays.
107.	★ Count Subarrays With Given AND.
108.	Maximum AND Pair in an Array.
109.	Maximum AND of Any Subset.
110.	Find Longest Subarray With Maximum AND.
111.	Find Subarray With Maximum OR.
112.	Minimum OR Required to Satisfy a Condition.

12. Bitmask Dynamic Programming

113.	★ Travelling Salesman Problem Using Bitmask DP.
114.	★ Assignment Problem Using Bitmask DP.
115.	Minimum Cost Perfect Matching.
116.	Hamiltonian Path Using Bitmask DP.
117.	Count Hamiltonian Paths.
118.	Minimum Cost to Visit All Nodes.
119.	Minimum Number of Operations Using Bitmask DP.
120.	Profile DP — Grid Tiling.
121.	DP Over Subsets.
122.	★ SOS DP — Sum Over Subsets.
123.	SOS DP — Maximum Over Subsets.
124.	Bitmask DP With Memoization.
125.	Bitmask DP With Tabulation.

13. Advanced Data Structures With Bitwise Operations

126.	★ Fenwick Tree With XOR.
127.	★ Segment Tree for Range XOR Queries.
128.	Segment Tree for Range AND/OR.
129.	Sparse Table for Range AND.
130.	Sparse Table for Range OR.
131.	Bitset-Based Subset Sum.
132.	Bitset Intersection and Union.
133.	Dynamic Bitset Operations.
134.	Range Update and Query Using Bitwise Techniques.

14. Advanced Interview and Competitive Programming Problems

135.	★ Maximum XOR of Two Numbers Under Constraints.
136.	Maximum XOR of a Subarray With Range Restrictions.
137.	★ Count Pairs With XOR Less Than K.
138.	Minimum Bit Flips to Satisfy an OR Condition.
139.	Minimum Bit Flips to Satisfy an AND Condition.
140.	Find Maximum Number After K Bit Changes.
141.	Find the Largest Number With Given Set Bits.
142.	Find the Smallest Number With Given Set Bits.
143.	Find Maximum Product Using Bitmasking.
144.	Meet in the Middle With Bitmasking.
145.	Coordinate Compression With Bitmask States.
146.	Bitwise Optimization for Subset Sum.
147.	Set Intersection Using Bitsets.
148.	Find Maximum AND/OR/XOR Under Given Constraints.

Recommended study order

Foundation: Sections 1–2
Core interview patterns: Sections 3–8
Advanced algorithms: Sections 9–12
Specialized data structures and hard variations: Sections 13–14

★ marks a high-priority pattern in this syllabus. It is a study recommendation, not a prediction of a company's exact interview questions.